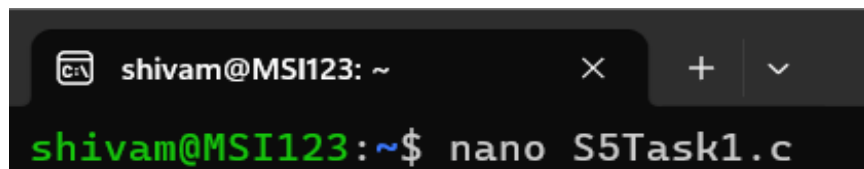


Week 5

Task 1

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency. Develop a program that executes the equivalent of the shell command: `ls-1 | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.



```
shivam@MSI123: ~  
shivam@MSI123:~$ nano S5Task1.c
```

```
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
#include <stdlib.h>  
#include <string.h>  
#include <time.h>  
int main() {  
    int pipefd[2];  
    pid_t pid;  
    char buffer[100];  
    int data[] = {10, 20, 30, 40, 50};  
    int n = 5;  
    if (pipe(pipefd) == -1) {  
        perror("pipe");  
        exit(1);  
    }  
    pid = fork();  
    if (pid < 0) {  
        perror("fork");  
        exit(1);  
    }  
    if (pid > 0) {  
        close(pipefd[0]);  
        printf("Producer (Parent):\n");  
        for (int i = 0; i < n; i++) {  
            sprintf(buffer, "%d", data[i]);  
            printf("Producing: %d\n", data[i]);  
            write(pipefd[1], buffer, strlen(buffer) + 1);  
            usleep(100000);  
        }  
    }  
}
```

```

        usleep(100000);
    }
    close(pipefd[1]);
    wait(NULL);
}
else {
    close(pipefd[1]);
    printf("\nConsumer (Child):\n");
    while (read(pipefd[0], buffer, sizeof(buffer)) > 0) {
        printf("Consumed: %s\n", buffer);
    }
    close(pipefd[0]);
}
return 0;
}
|

```

OUTPUT:

```

shivam@MSI123:~$ nano S5Task1.c
shivam@MSI123:~$ gcc S5Task1.c -o S5Task1
shivam@MSI123:~$ ./S5Task1
Producer (Parent):
Producing: 10

Consumer (Child):
Consumed: 10
Producing: 20
Consumed: 20
Producing: 30
Consumed: 30
Producing: 40
Consumed: 40
Producing: 50
Consumed: 50
shivam@MSI123:~$ |

```

Task 2

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
shivam@MSI123:~$ nano S5Task2.c
shivam@MSI123:~$ |
```

```
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    int pipefd[2];
    pid_t pid1, pid2;
    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }
    // Create first child
    pid1 = fork();
    if (pid1 == -1)
    {
        perror("fork");
        exit(1);
    }
    if (pid1 == 0)
    {
        // First child: ls -l
        close(pipefd[0]);
        // Redirect stdout to pipe
        dup2(pipefd[1], STDOUT_FILENO);
        close(pipefd[1]);
        execlp("ls", "ls", "-l", (char *)NULL);
        perror("execlp ls");
    }
}
```

```

// Create second child
pid2 = fork();
if (pid2 == -1)
{
    perror("fork");
    exit(1);
}
if (pid2 == 0)
{
    // Second child: grep ".c"
    close(pipefd[1]);
    // Redirect stdin from pipe
    dup2(pipefd[0], STDIN_FILENO);
    close(pipefd[0]);
    execlp("grep", "grep", ".c", (char *)NULL);
    perror("execlp grep");
    exit(1);
}
// Parent process
close(pipefd[0]);
close(pipefd[1]);
waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);
printf("Command executed successfully.\n");
return 0;
}

```

OUTPUT:

```
shivam@MSI123:~$ nano S5Task2.c
shivam@MSI123:~$ gcc S5Task2.c -o S5Task2
shivam@MSI123:~$ ./S5Task2
-rw-r--r-- 1 shivam shivam 1041 Sep 11 16:27 S5Task1.c
-rw-r--r-- 1 shivam shivam 1076 Sep 11 16:38 S5Task2.c
-rwxr-xr-x 1 shivam shivam 16176 Aug 4 08:23 copy
-rw-r--r-- 1 shivam shivam 904 Aug 4 08:13 copy.c
-rw-r--r-- 1 shivam shivam 900 Aug 1 03:40 nan.c
-rw-r--r-- 1 shivam shivam 964 Sep 11 09:18 ptask1.c
-rw-r--r-- 1 shivam shivam 1012 Sep 11 09:25 ptask2.c
-rw-r--r-- 1 shivam shivam 440 Aug 4 06:45 shell.c
-rw-r--r-- 1 shivam shivam 2947 Aug 19 16:02 shell_parser.c
-rw-r--r-- 1 shivam shivam 3705 Aug 19 16:17 task2.c
-rw-r--r-- 1 shivam shivam 2945 Aug 20 04:51 task3.c
-rw----- 1 shivam shivam 2944 Aug 20 04:49 task3.c.save
-rw----- 1 shivam shivam 1 Aug 20 04:50 task3.c.save.1
-rw-r--r-- 1 shivam shivam 4466 Aug 11 07:05 week4.c
-rw-r--r-- 1 shivam shivam 1 Aug 11 07:09 week4.c.save
Command executed successfully.
shivam@MSI123:~$ |
```